Some Common Dose-Exposure-Response Estimands and Conditions for their Causal Identifiability

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APPENDIX

Generation of the simulated example data

The example dataset is constructed proceeding in the following steps:

- Covariate values sampled from parametric univariate distributions that approximately matched the reported summary statistics in the published demographic tables. This method of covariate value simulation is not expected to recapitulate the multivariate dependence between the covariates that would be present in the real data.
- Doses were assigned to conform to the sample size distribution in the original studies.
- Exposure metrics were simulated using a simplified version of the published popPK model, with parameter values similar to the reported estimates.
- Exposure-response (E-R) was simulated using a logistic regression model with parameter values similar to the reported estimates.

Summary of simulated example data

The distribution of demographic / baseline variables is similar to the study reported in literature (see below Table S1). There were 77 individuals in the virtual J101 study, S=1, and only 14 of them were adult T-cell leukemia/lymphoma (ATLL) patients. All 25 individuals in the virtual J201 study, S=2, were ATLL patients. The individuals in the S=2 study were all Asians, of older age, higher alpha-1-acid glycoprotein (AAG) and lower body weight compared to the individuals in S=1. Creatinine clearance (CL) was generated from a hypothetical linear model involving age, gender and baseline lactate dehydrogenase concentration (LDH) as it was not reported in the literature. There were 24.7% and 32% of individuals with Creatinine CL <= 60 mL/min in S=1 and S=2, respectively.

Patients for the target trial S=0 were constructed using the observed covariate values in the ATLL subset of the combined data from S=1 and S=2. (Computationally this amounts to re-using the same patients in different studies, but notionally the patients in S = 0 should still be conceived of as “different patients”.) The S = 00 population was simply taken to be the subset of S = 0 with Creatinine CL <= 60 mL/min.

Table S1. Summary statistics on studies

STUDY	Virtual J101, S=1	Virtual J201, S=2	Target Trial, S=0 (ATLL patients in J101 and J201)
N	77	25	39
Age, years			
Median	68	71	70
Mean (Std. Dev.)	67.4 (10.9)	70.6 (4.5)	68.8 (8.3)
Range (Min; Max)	(36; 89)	(63; 81)	(48; 89)

AAG, mg/dL			
Median	87.1	98	95.1
Mean (Std. Dev.)	88.15 (32.06)	98.49 (43.52)	94.49 (40.97)
Range (Min; Max)	(4.0; 165.5)	(18.2; 197.1)	(18.2; 197.1)
Body weight, kg			
Median	67	59	62
Mean (Std. Dev.)	67.6 (13.7)	58.2 (11.9)	63.9 (15.6)
Range (Min; Max)	(23; 103)	(32; 78)	(32; 103)
LDH, U/L			
Median	259.3	276.1	271.3
Mean (Std. Dev.)	383.75 (317.46)	495.43 (332.71)	458.75 (378.69)
Range (Min; Max)	(117.0; 1960.2)	(149.8; 1320.9)	(149.8; 1960.2)
Hb, g/L			
Median	119.1	112.4	116.3
Mean (Std. Dev.)	117.72 (18.98)	113.34 (18.15)	114.95 (18.64)
Range (Min; Max)	(78.8; 170.6)	(88.3; 152.1)	(78.8; 152.1)
PLT, 10^9 /L			
Median	196.6	183.8	198.4
Mean (Std. Dev.)	197.00 (77.93)	239.19 (158.72)	240.72 (138.84)
Range (Min; Max)	(26.5; 411.1)	(79.6; 620.1)	(77.7; 620.1)
ANC, 10^9 /L			
Median	3.9	5.5	5.4
Mean (Std. Dev.)	3.90 (1.33)	5.80 (2.83)	5.41 (2.48)
Range (Min; Max)	(0.4; 6.8)	(0.9; 10.9)	(0.9; 10.9)
Creatinine Clearance, mL/min			
Median	73.4	71.6	74.8
Mean (Std. Dev.)	67.48 (18.50)	68.05 (21.63)	68.52 (19.78)
Range (Min; Max)	(14.9; 98.0)	(17.2; 96.5)	(14.9; 96.5)
Sex; N(%)			
Male	43 (55.8%)	12 (48.0%)	21 (53.8%)
Female	34 (44.2%)	13 (52.0%)	18 (46.2%)
ECOG PS score; N(%)			
0	40 (51.9%)	13 (52.0%)	17 (43.6%)
>=1	37 (48.1%)	12 (48.0%)	22 (56.4%)
Hepatic function; N(%)			
Normal	65 (84.4%)	17 (68.0%)	26 (66.7%)
Mild	12 (15.6%)	8 (32.0%)	13 (33.3%)
Race; N(%)			
Asian	46 (59.7%)	25 (100.0%)	33 (84.6%)
White	23 (29.9%)	0 (0.0%)	2 (5.1%)
Black	8 (10.4%)	0 (0.0%)	4 (10.3%)
Location; N(%)			

Japan	46 (59.7%)	25 (100.0%)	33 (84.6%)
USA	31 (40.3%)	0 (0.0%)	6 (15.4%)
Creatinine Clearance Level; N(%)			
>60 mL/min	58 (75.3%)	17 (68.0%)	30 (76.9%)
<=60 mL/min	19 (24.7%)	8 (32.0%)	9 (23.1%)
Patient Type; N(%)			
Non-ATLL	63 (81.8%)	0 (0.0%)	0 (0.0%)
ATLL	14 (18.2%)	25 (100.0%)	39 (100.0%)
Treatment; N(%)			
150 mg	7 (9.1%)	0 (0.0%)	2 (5.1%)
200 mg	61 (79.2%)	25 (100.0%)	33 (84.6%)
250 mg	7 (9.1%)	0 (0.0%)	3 (7.7%)
300 mg	2 (2.6%)	0 (0.0%)	1 (2.6%)

Abbreviations: AAG, alpha-1-acid glycoprotein; ATLL, adult T-cell leukemia/lymphoma; LDH, lactate dehydrogenase concentration; Hb, hemoglobin; PLT, platelet count; ANC, absolute neutrophil count; ECOG PS, Eastern Cooperative Oncology Group performance status.

Supplementary results

Cohort-based Summaries

Cohort-based summaries targeting treatment policy estimands of the studies, e.g., $\Pr(R^d|S=1)$ may serve as reference for validation of the D-R and D-E-R estimands. Estimates obtained with cohort based summaries should be unbiased due to random selection of study subjects (see also section on estimands and discussion on unobserved confounding). The cohort-based summaries can be compared to D-R and D-E-R estimates using the same models as used for the estimands of clinical interest. The D-R or D-E-R estimates could be biased due to unobserved confounding (see discussion section on unobserved confounding). An important difference between the cohort based summaries and the D-R and D-E-R summaries would be an indication for unobserved confounding. This validation approach has the limitation that in pooled analyses the cohorts of interest may be very small, limiting the precision of the cohort based summaries. In our example, there was no unobserved confounding and most cohorts were small.

The cohort-based summaries may also be seen as estimates of the estimands of clinical interest, $\Pr(R^d|S=0)$, if it is assumed that the population in the cohorts approximates the target population (Table S2). In our example, these estimates were of limited quality and insufficient to support assessment of the dose-response. In contrast to the D-R and D-E-R analyses, the cohort-based summary approach (observed group mean and adjusted group mean) did not capture the positive dose-response relationship (the estimates for 300 mg were lower than 250 mg). This was because there were very limited ATLL patients assigned to these two groups and the non-ATLL patients in these two groups were not representative samples of the target population (ATLL). Additionally, the cohort-based summary estimates for 200 mg group using Study S=2 were also biased even though the 25 subjects were all ATLL patients. This was because these 25 patients were all from the same location and with

low demographic diversity -- still not a representative sample for ATLL population. Adjusted group mean (for ATLL population or for ATLL with CRCL <= 60 population) was even more biased.

Table S2. Simulation results for cohort based summaries

	True Incidence Rate	Estimation		Bias		MSE	
		Obs Group Mean*	Adj Group Mean**	Obs Group Mean*	Adj Group Mean**	Obs Group Mean*	Adj Group Mean**
Estimand 1'							
150 mg	0.1452	0.1760	0.1741	0.0308	0.0289	0.0745	0.0306
200 mg	0.1876	0.1479	0.1386	-0.0397	-0.0490	0.0167	0.0125
		0.2077 [a]	0.2077 [a]	0.0201 [a]	0.0201 [a]	0.0062 [a]	0.0062 [a]
250 mg	0.2378	0.1753	0.2133	-0.0625	-0.0246	0.0521	0.0312
300 mg	0.2947	0.2400	0.2028	-0.0547	-0.0919	0.1854	0.0922
Estimand 2'							
150 mg	0.2414	0.1080	0.1689	-0.1334	-0.0726	0.0651	0.0222
200 mg	0.2963	0.1830	0.1810	-0.1133	-0.1153	0.0259	0.0234
		0.3403 [a]	0.3403 [a]	0.0440 [a]	0.0440 [a]	0.0085 [a]	0.0085 [a]
250 mg	0.3538	0.3940	0.4001	0.0402	0.0463	0.0702	0.0421
300 mg	0.4116	0.4750	0.3526	0.0634	-0.0590	0.2534	0.1195
Estimand 3' (CRCL <= 60)							
150 mg	0.1980	-	0.2303	-	0.0322	-	0.0584
200 mg	0.2584	0.2000	0.1960	-0.0584	-0.0623	0.1634	0.0299
		0.2690 [a]	0.2690 [a]	0.0106 [a]	0.0106 [a]	0.0220 [a]	0.0220 [a]
250 mg	0.3274	-	0.2744	-	-0.0530	-	0.0597
300 mg	0.4016	-	0.2284	-	-0.1731	-	0.1395

*Observed treatment group incidence rate of target population in each study without adjustment.

** Adjusted treatment group incidence rate of target population in each study. Patient type and CRCL status were included in the adjustment.

- No observations available.

[a] Corresponding to $\Pr(R^d | S = 2, P = ATLL)$. The rest statistics correspond to $\Pr(R^d | S = 1, P = ATLL)$.

Sensitivity analyses

Sensitivity analyses were implemented to show that the comparative performance of the different estimation methods depends on the particularities of the pooled study data. When increasing the error of the exposure, MSE of the D-E-R estimates increases (Table S3). For the chosen increase in error by a factor of lognormal(0,0.15), there was a moderate increase of the MSE by a few percent largely consistent across all estimates.

When failing to adjust for important covariates that affect differential response in the different trials of the pool, bias increases irrespective of whether D-R or D-E-R analyses are used. This was shown by excluding sex and race from the covariates, L_E and L_Y, used for adjustment in the analysis. Sex and Race were unbalanced in the different studies (Table S1) and did affect exposure and outcome. Excluding them from the analysis corresponds to

having unobserved confounding and leads to an increase in bias for most of the estimates (Table S4). This unobserved confounding affects both the D-R and the D-E-R estimates.

Table S3: Sensitivity Analysis: Simulation results for estimands of clinical interest with additional error applied to exposure from different studies

	True Incidence Rate	Estimate		Bias		MSE	
		Additional error applied to J101	Additional error applied to J201	Additional error applied to J101	Additional error applied to J201	Additional error applied to J101	Additional error applied to J201
Estimand 1: Dose reduction due to TEAE for ATLL patients							
150 mg	0.1452	0.1567	0.1516	0.0115	0.0064	0.0027	0.0025
200 mg	0.1876	0.1855	0.1858	-0.0020	-0.0017	0.0019	0.0019
250 mg	0.2378	0.2483	0.2473	0.0105	0.0094	0.0064	0.0059
300 mg	0.2947	0.3224	0.3200	0.0277	0.0253	0.0173	0.0162
Estimand 2: Grade >= 3 PLT decrease for ATLL patients							
150 mg	0.2414	0.2466	0.2418	0.0051	0.0003	0.0028	0.0028
200 mg	0.2963	0.2933	0.2936	-0.0030	-0.0027	0.0018	0.0018
250 mg	0.3538	0.3580	0.3582	0.0042	0.0044	0.0049	0.0044
300 mg	0.4116	0.4209	0.4203	0.0092	0.0087	0.0114	0.0103
Estimand 3 Dose reduction due to TEAE for ATLL patients with CRCL <= 60							
150 mg	0.1980	0.2329	0.2254	0.0349	0.0273	0.0174	0.0144
200 mg	0.2584	0.2552	0.2559	-0.0031	-0.0024	0.0087	0.0088
250 mg	0.3274	0.3246	0.3178	-0.0028	-0.0096	0.0281	0.0240
300 mg	0.4016	0.3891	0.3756	-0.0124	-0.0260	0.0596	0.0516

Legend: Assessment of the estimated outcomes of the 1000 simulated trials. True incidence rate, known from the setup of the simulated trials; summaries of the estimates as mean (estimate), bias, and mean square error (MSE).
Note: Building on the same virtual dataset used in Table 2, AUCSS used in the simulations for the response was made to differ from the measured AUCSS by multiplying it with an error term (lognormal(0,0.15)). This was done either for study J101 or study J201. All other analysis settings remained unchanged.

Table S4: Sensitivity Analysis: Simulation results for estimands of clinical interest with unmeasured confounded covariates (Sex and Race)

	True Incidence Rate	Estimate		Bias		MSE	
		D-E-R	D-R	D-E-R	D-R	D-E-R	D-R
Estimand 1: Dose reduction due to TEAE for ATLL patients							
150 mg	0.1452	0.1465	0.1914	0.0014	0.0462	0.0024	0.0142
200 mg	0.1876	0.1883	0.1863	0.0007	-0.0012	0.0017	0.0019
250 mg	0.2378	0.2522	0.2158	0.0143	-0.0220	0.0054	0.0116
300 mg	0.2947	0.3238	0.2661	0.0291	-0.0285	0.0152	0.0371
Estimand 2: Grade >= 3 PLT decrease for ATLL patients							
150 mg	0.2414	0.2399	0.2640	-0.0015	0.0225	0.0028	0.0084
200 mg	0.2963	0.2967	0.2964	0.0004	0.0001	0.0016	0.0019
250 mg	0.3538	0.3633	0.3380	0.0096	-0.0158	0.0038	0.0091
300 mg	0.4116	0.4271	0.3803	0.0155	-0.0313	0.0091	0.0278
Estimand 3 Dose reduction due to TEAE for ATLL patients with CRCL <= 60							
150 mg	0.1980	0.1911	0.2551	-0.0069	0.0570	0.0086	0.0258
200 mg	0.2584	0.2573	0.2526	-0.0010	-0.0057	0.0083	0.0092
250 mg	0.3274	0.3639	0.2822	0.0365	-0.0452	0.0250	0.0239
300 mg	0.4016	0.4674	0.3285	0.0658	-0.0731	0.0537	0.0553

Legend: Assessment of the estimated outcomes of the 1000 simulated trials. True incidence rate, known from the setup of the simulated trials; summaries of the estimates as mean (estimate), bias, and mean square error (MSE).
Note: The virtual dataset and analysis method used in this simulation study are identical to those described for Table 2. The only difference is that, in this scenario, sex and race are treated as unobserved variables, thereby introducing them as unobserved confounders.

Examples of model diagnostics

The section shows some possible model diagnostics for one of the simulated trials.

D-E-R Analysis

Figure S1: Dose-Exposure Model Diagnostics

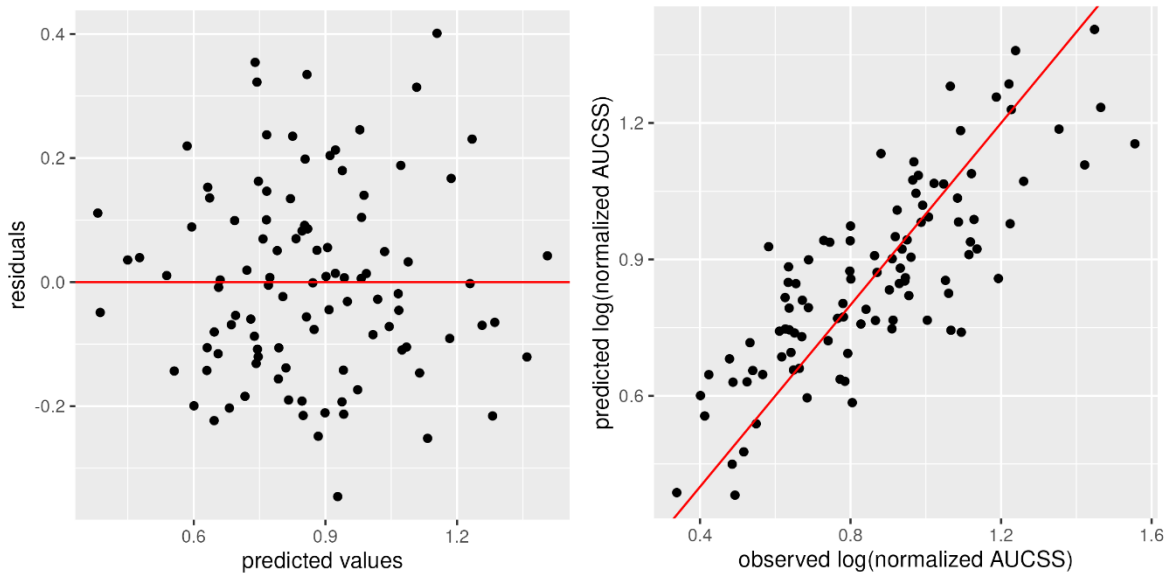
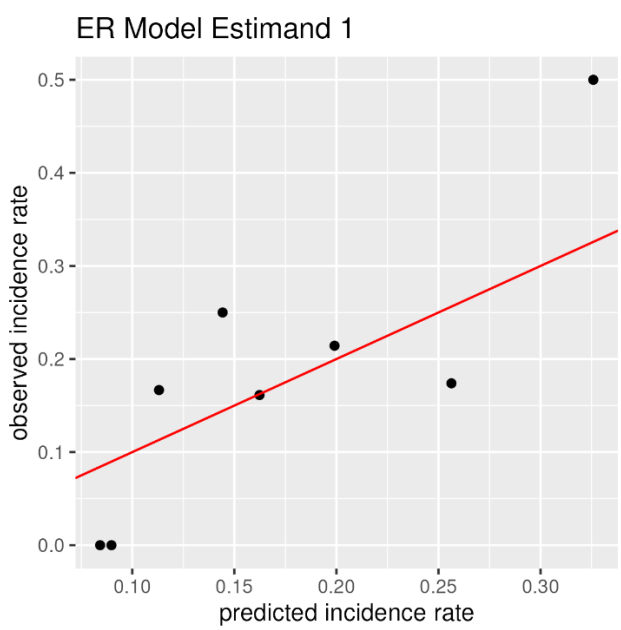


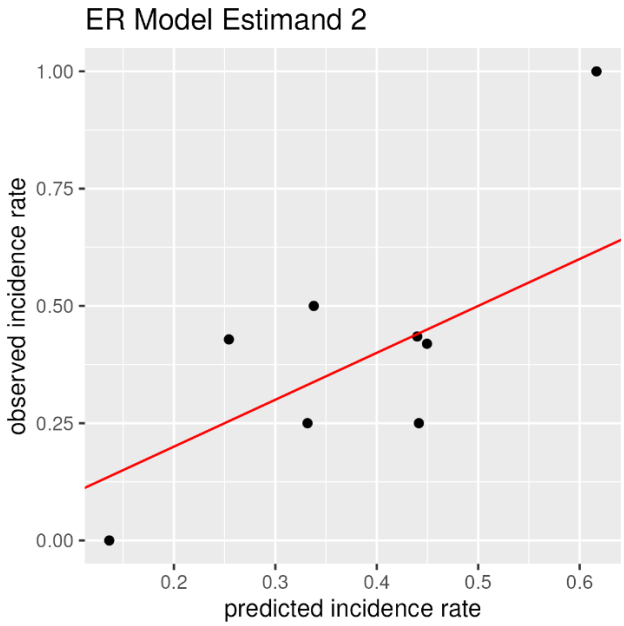
Figure S2: Exposure-Response Model Diagnostics (Estimand 1)




```
Exposure-Response Model Estimand 1

    Hosmer and Lemeshow goodness of fit (GOF) test

data:  AE.dt1$DV, AE.dt1$pred
X-squared = 5.4918, df = 8, p-value = 0.7039
```

Left: comparison of the model-predicted incidence rates and the observed incidence rates. Study participants were grouped into 8 bins based on their observed exposure level (AUCSS).
Right: H-L test results.

Figure S3: Exposure-Response Model Diagnostics (Estimand 2)

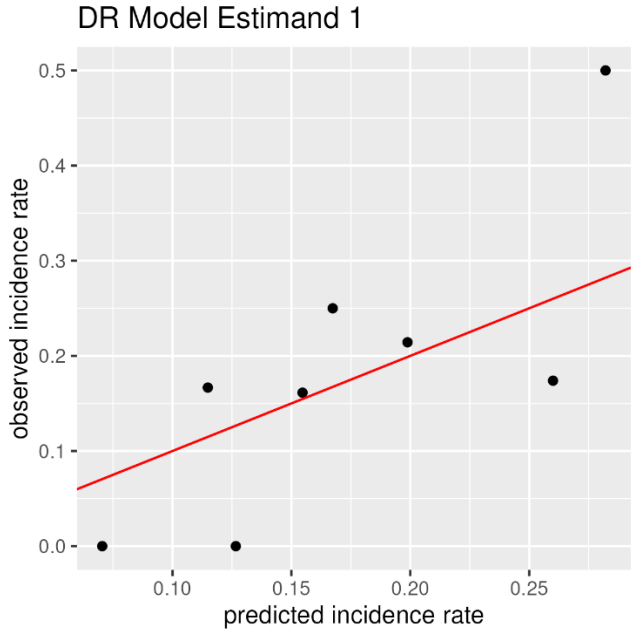



```
Exposure-Response Model Estimand 2

    Hosmer and Lemeshow goodness of fit (GOF) test

data:  AE.dt2$DV, AE.dt2$pred
X-squared = 6.5292, df = 8, p-value = 0.5882
```

Left: comparison of the model-predicted incidence rates and the observed incidence rates. Study participants were grouped into 8 bins based on their observed exposure level (AUCSS).
Right: H-L test results.

D-R Analysis

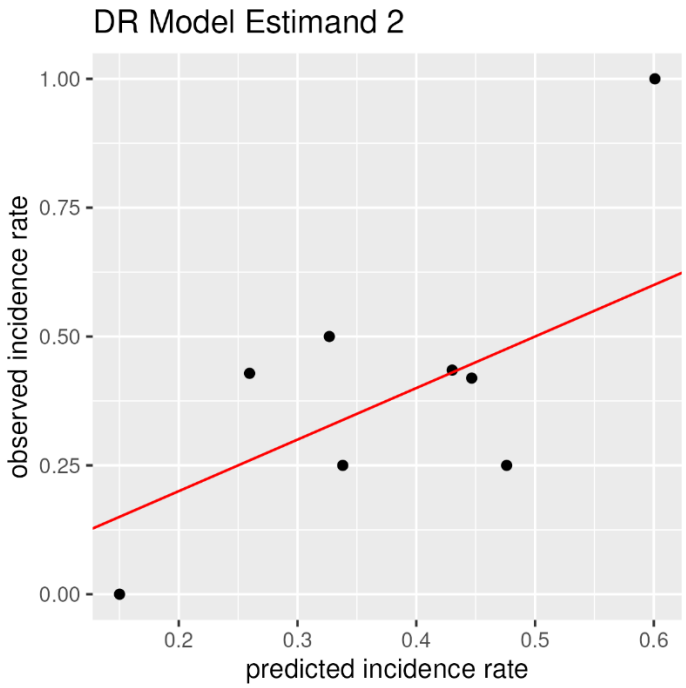
Figure S4: Dose-Response Model Diagnostics (Estimand 1)




```
Dose-Response Model Estimand 1

    Hosmer and Lemeshow goodness of fit (GOF) test

data:  DR.dt1$DV, DR.dt1$pred
X-squared = 12.563, df = 8, p-value = 0.1278
```

Left: comparison of the model-predicted incidence rates and the observed incidence rates. Study participants were grouped into 8 bins based on their observed exposure level (AUCSS).
Right: H-L test results.

Figure S5: Dose-Response Model Diagnostics (Estimand 2)




```
Dose-Response Model Estimand 2

    Hosmer and Lemeshow goodness of fit (GOF) test

data:  DR.dt2$DV, DR.dt2$pred
X-squared = 3.2808, df = 8, p-value = 0.9155
```

Left: comparison of the model-predicted incidence rates and the observed incidence rates. Study participants were grouped into 8 bins based on their observed exposure level (AUCSS).
Right: H-L test results.